Probability & Statistics

Compound Event → $P(A \cap B) = P(A) \cdot P(B) = P(A) \cdot P(B/A)$

- $\cap$ and $\times$ (compound event)
- $\cup$ or $+$ (mutually exclusive event).

Q→ A box contain 3 white & 3 black ball. Another box contain 2 white & 4 black ball. On box is selected at random then one ball is drawn from the selected box. What is the P(E), the drawn ball is white?

$$P(W) = P(W \cap I) \cup P(W \cap II)$$
$$= P(I) \cdot P(W/I) + P(II) \cdot P(W/II)$$
$$= \frac{1}{2} \cdot \frac{3}{6} + \frac{1}{2} \cdot \frac{2}{6} = 5/12.$$

Q→ A box contain 3 white & 4 black balls. 2 balls are selected at random.
(i) P(E) that both balls are white? → ${}^3C_2 / {}^7C_2 = 1/7$
(ii) P(E) that I ball is white & II is black?
(iii) both balls are opp. colour? → $P(E) = \frac{n(E)}{n(S)} = \frac{{}^3C_1 \times {}^4C_1}{{}^7C_2} = 12\left(\frac{3}{7} \times \frac{4}{6}\right)$

(ii) $P(E) = \frac{3}{7} \times \frac{4}{6} = 2/7$

Q→ A box contain 3 white & 4 black ball. Another box contain 2 white & 5 black ball. One ball is transferred from box I to box II then a ball is drawn from the II box. P(E) the drawn ball is white?

Case I: When white ball is transferred.
$= \frac{{}^3C_1}{{}^7C_1} = \frac{3}{7}$
P(E) of drawing ball $= \frac{3}{8}$

Case II: black transferred.
$= \frac{4}{7}$
P(E) of drawing $= \frac{2}{8}$

$$P(E) = \frac{3}{7} \times \frac{3}{8} + \frac{4}{7} \times \frac{2}{8} = \frac{17}{56}$$

Q→ 3 group of children contain 3 girls & 1 boy, 2G & 2B, 1G & 3B resply. 1 child is selected at random from each grp. Find the chance that 3 selected comprise 1G & 2B.

1G 2B →
G B B = $3/4 \times 2/4 \times 3/4 = 18/64$
B G B = $1/4 \times 2/4 \times 3/4 = 6/64$
B B G = $1/4 \times 2/4 \times 1/4 = 2/64$

$$P(E) = 18/64 + 6/64 + 2/64 = 26/64 = 13/32$$

Q→ 'A' has 3 shares in a lottery containing 3 prize & 9 blanks. 'B' has 2 share in a lottery containing 2 prize & 6 blanks. Compare their chance of winning?

(chance of winning = atleast 1 prize winning)

$P(\bar{A}) = \frac{{}^9C_3}{{}^{12}C_3}$ $P(A) = 1 - \frac{{}^9C_3}{{}^{12}C_3}$ $P(\bar{B}) = \frac{{}^6C_2}{{}^8C_2}$ $P(B) = 1 - \frac{{}^6C_2}{{}^8C_2}$

$P(A) = \frac{\overset{3}{9}\times 8\times 7}{12\times 11\times \overset{3}{10}_5} = 1 - \frac{21}{55} = \frac{34}{55}$.

$P(B) = 1 - \frac{{}^{9}C_2}{{}^{8}C_2} = 1 - \frac{\overset{3}{8}\times 5}{4\times 7} = \frac{13}{28}$

$\boxed{P(A) > P(B)}$

Q→ If success of the event & P(E) is defined by 'p'

A	$\frac{1}{q}$	$3pq^2$	$5q^4p$	$7 \ldots \infty$
B	2	4	6	
	$2\cdot q$			

• P(A) winning when A has started the game.

$P(A) = P(1) \text{ or } P(3) \text{ or } P(5) \text{ or } P(7) \text{ or } \ldots \infty$

$P(A) = p + q^2p + q^4\cdot p + q^6\cdot p + \ldots \infty$

$P(A) = p(1 + q^2 + q^4 + q^6 + \ldots \infty)$ → Infinite G.P series $\left(\frac{1}{1-x}\right)$.

$P(A) = p\left(\frac{1}{1-q^2}\right) = \frac{p}{1-q^2}$.

$P(B) = P(2) \text{ or } P(4) \text{ or } P(6) \text{ or } \ldots \infty$

$P(B) = q\cdot p + q^3\cdot p + q^5\cdot p + \ldots \infty$

$P(B) = qp(1 + q^2 + q^4 + \ldots \infty)$

$P(B) = \frac{pq}{1-q^2}$ ∴ P(B) when A has started the game.

$\Rightarrow \boxed{P(A) > P(B)}$

★ A, B, C, D are 4 persons and A got the first chance.

A	1	5	9	13
B				
C				
D				

$P(A) = P(1) \text{ or } P(5) \text{ or } P(9) \text{ or } P(13) \text{ or } \ldots \infty$

$P(A) = p + q^4\cdot p + q^8\cdot p + q^{12}\cdot p + \ldots \infty$

$P(A) = P\left(\frac{1}{1-q^4}\right) = \frac{p}{1-q^4}$

$P(B) = \frac{pq}{1-q^4}$ $\quad P(C) = \frac{q^2\cdot p}{1-q^4}$

$P(D) = \frac{q^3\cdot p}{1-q^4}$

Q→ A and B throw a coin alternative till one of them gets a head & win the game. Find their respective P(E) of winning.

$p = P(H) = 1/2$ $\quad q = 1/2$

$P(A) = \frac{p}{1-q^2} = \frac{1/2}{1-1/4} = \frac{1/2}{3/4} = 2/3.$

$P(B) = \frac{1/2\cdot 1/2}{1-1/4} = 1/3.$

Q→ 2 coin are there. Success is getting 2 heads.

$p = 1/4 \quad q = 3/4 \quad P(A) = \frac{p}{1-q^2} = \frac{1/4}{1-\left(\frac{3}{4}\right)^2} = \frac{1/4}{1-\frac{9}{16}} = \frac{1/4}{7/16} = \frac{4}{7}$

BAYE'S THEOREM

Q→ A box contain 3 white & 3 black balls. Another box has 2 white & 4 black one box is selected at random then one ball is drawn from the selected box. If the drawn ball is white, what is P(E), it was from the I box? $P\left(\frac{B_1}{A}\right) = \frac{P\left(\frac{A}{B_1}\right)\cdot P(B_1)}{P(A)} = \frac{P\left(\frac{A}{B_1}\right)\cdot P(B_1)}{P\left(\frac{A}{B_1}\right)\cdot P(B_1) + P\left(\frac{A}{B_2}\right)\cdot P(B_2)}$

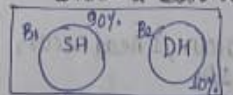
$= \frac{\frac{3}{6}\cdot\frac{1}{2}}{\frac{3}{6}\cdot\frac{1}{2} + \frac{2}{6}\cdot\frac{1}{2}} = \frac{\frac{3}{12}}{\frac{5}{12}} = \frac{3}{5}$

Baye's Theorem Formula:

If $B_1, B_2, \ldots, B_n$ are mutually exclusive event. Let 'A' is an event which occur from B_1 or B_2 or $\ldots B_n$.

$$\boxed{P\left(\frac{B_i}{A}\right) = \frac{P\left(\frac{A}{B_i}\right)\cdot P(B_i)}{P(A)} \qquad P(A) = \sum P\left(\frac{A}{B_i}\right)\cdot P(B_i) \quad i = 1, 2, \ldots, n.}$$

Q→ 2 types of coins — one single head & one double head in a box. 10% coins are double head. Now 2 coins are selected at random and tossed & both occur head. P(E) that both coins are single headed?



$P(B_1) = 9/10 \quad P(B_2) = 1/10$

$P(A/B_1) = 1/4 \quad P(A/B_2) = 1$

$P(A) = P(2H)$

$P(B_1/A) = \frac{P(A/B_1)\cdot P(B_1)}{P(A)}$

$P(B_1/A) = \frac{1/4\cdot 9/10}{1/4\cdot 9/10 + 1\cdot 1/10}$

$\boxed{P(B_1/A) = 9/13.}$

Q. In a factory which manufactures bolts, machine A, B, C manufacture respectively 25%, 35% & 40% of the total bolt. Of their outputs, 5%, 4% and 2% are defective bolts. A bolt is drawn at random from the product and is found to be defective. P(E) it is manufactured by (i) Machine A (ii) Machine B or machine C.

(i) $P(A)=\frac{25}{100}$ $P(B)=\frac{35}{100}$ $P(C)=\frac{40}{100}$

$P(A_1)=\frac{5}{100}$ $P(B_1)=\frac{4}{100}$ $P(C_1)=\frac{2}{100}$

event E = defective bolt.

$$P\left(\frac{B_1}{E}\right)=\frac{P\left(\frac{E}{B_1}\right)\cdot P(B_1)}{P(E)}$$

$$P\left(\frac{B_1}{E}\right)=\frac{\frac{25}{100}\times\frac{5}{100}}{\frac{25}{100}\times\frac{5}{100}+\frac{35}{100}\times\frac{4}{100}+\frac{40}{100}\times\frac{2}{100}}=\frac{125}{125+140+80}$$

$$P\left(\frac{B_1}{E}\right)=\frac{125}{345}=\frac{25}{69}$$

(ii) E = bolt from machine B or machine C.

$$P\left(\frac{B_2}{E}\right)+P\left(\frac{B_3}{E}\right)=\frac{140}{345}+\frac{80}{345}=\frac{220}{345}=\frac{44}{69}$$

or $1-P\left(\frac{B_1}{E}\right)=1-\frac{25}{69}=\frac{44}{69}$.

* Binomial Distribution.

If 'p' is a probability of success in single trial then in 'n' trial, the probability of rth success is defined as —

$$P(r)={}^nC_r\,p^r q^{n-r} \quad ; r=0,1,2,\ldots,n$$

We use it when n = small, p = large.

$n \le 30$

Q. 2 coins are tossed then find P(E) of 0 head occur, 1 head occur, 2 head occur. $n=2$

$P(0H)=P(TT)=1/4$

$P(1H)=P(HT)\text{ or }P(TH)=2/4$

$P(2H)=P(HH)=1/4$

Q. 10 coins are tossed then P(E) of 6 Heads?

$n=10$ $P(H)=1/2$ $P(T)=1/2$. $r=6$

$$P(r)={}^nC_r p^r q^{n-r}$$

$$P(6)={}^{10}C_6\left(\frac{1}{2}\right)^6\left(\frac{1}{2}\right)^4=\frac{10\times9\times8\times7}{24}\left(\frac{1}{2}\right)^{10}=\frac{210}{2^{10}}=\frac{105}{2^9}.$$

Q. Two coins are tossed 16 times then how many (frequency) times 0H occurs, 1H occur, 2H occur?

No. of frequency = 16, $n=2$. → N, n do the milenge.

$P(H)=1/2$ (single trial), $q=1/2$

n = No. of Heads = 0, 1, 2.

~~Q. 10 coins are tossed~~

($n=2$) ($N=16$)

x	0 (TT)	1 (HT, TH)	2 (HH)	
P(x)	1/4	2/4	1/4	$\rightarrow \Sigma P(x)=1$
fe(x) = P(x)·N	4	8	4	$\Sigma fe(x)=N$

$$\text{Mean}=\frac{\Sigma f\cdot x}{\Sigma f=N}=\frac{(0\times4)+(1\times8)+(2\times4)}{16}=1.$$

$\bar{x}=\text{mean}=E(x)=\Sigma x\cdot P(x)$

$\sigma^2=\Sigma(x-\bar{x})^2\cdot E(x)$.

- Mean of binomial distribution is ~~n*p~~ $p\times n$.
- Variance of binomial distribution is $n\times p\times q$

*. To show that mean of the binomial distribution is np.

$$E(X)=\sum_{x=0}^{n} x\cdot p(x)=\sum_{x=0}^{n} x\cdot {}^nC_x p^x q^{n-x}$$

$$=\sum x\frac{\lfloor n}{\lfloor n-x\rfloor x}p^x q^{n-x}$$

$$=\sum x\frac{\lfloor n}{x\lfloor n-x\rfloor x-1}p^x\cdot q^{n-x}$$

$$=\sum\frac{\lfloor n}{\lfloor x-1\rfloor n-x}p^x\cdot q^{n-x}$$

$$=\sum_{x=1}^{n}\frac{n\lfloor n-1}{\lfloor x-1\rfloor n-x}p^x\cdot q^{n-x}$$

$$=np\sum_{x=1}^{n}\frac{\lfloor n-1}{\lfloor x-1\rfloor(n-1)-(x-1)}\cdot p^{x-1}\cdot q^{(n-1)-(x-1)}$$

$\therefore \sum_{r=0}^{n} {}^nC_r p^r q^{n-r}=(q+p)^n$.

$= np(q+p)^{n-1}$

$= np(1)^{n-1} = np$ proved

* To show that variance of binomial distribution is npq.

$$\sigma^2 = \sum_{x=0}^{n} (x-\bar{x})^2 p(x)$$

or

$$\boxed{\sigma^2 = \Sigma x^2 p(x) - (\Sigma x \cdot p(x))^2}$$

$$\boxed{\sigma^2 = \Sigma x^2 p(x) - (np)^2} \quad -①$$

Putting $x^2 = x(x-1)+x$

$$\Sigma x^2 \cdot p(x) = \Sigma (x(x-1)+x)\, p(x)$$
$$= \Sigma x(x-1) \cdot P(x) + \Sigma x p(x)$$
$$= \Sigma x(x-1) \cdot P(x) + np$$
$$\sigma^2 = [n(n-1)\cdot p^2 + np] - (np)^2$$
$$= n^2p^2 - np^2 + np - n^2p^2$$
$$= -np^2 + np$$
$$= np(1-p)$$
$$= npq$$

Q→ In a precision bombing attack, there is a 50% chance that any one bomb will strike the target. 2 direct hits are required to destroy a target completely. How many bomb must be dropped such that a target destroy 99% or better?

$p = 1/2$ $q = 1/2$ n = no. of bomb. $x = 2, 3, 4, \ldots, n$

$P(x) = {}^nC_x\, p^x q^{n-x}$

$P(0) = {}^nC_0\, p^0 q^{n-0} = \left(\frac{1}{2}\right)^n$

$P(1) = {}^nC_1\, p^1 q^{n-1} = \frac{n}{2^n}$

$P(x) \geq 2$

$P(2) + P(3) + \ldots + P(n) \geq 0.99$

$P(x \geq 2) = 1 - [P(0) + P(1)]$

$P(x \geq 2) = 1 - \left[\frac{1}{2^n} + \frac{n}{2^n}\right] \geq 0.99$

$= 1 - \left[\frac{1+n}{2^n}\right] \geq 1 - 0.01$

$\frac{n+1}{2^n} \leq 0.01$

$\boxed{100n + 100 \leq 2^n}$

If $n=2$, $200+100 \leq 2^2$ ✗

$n=5$, $500+100 \leq 2^5$ ✗

$n=10$, $1000+100 \leq 2^{10}$ ✗

$n=11$, $1100+100 \leq 2^{11}$ ✓

minimum no. of bombs required = 11

Mode of the Binomial Distribution.

the size of the data whose frequency is maximum.

Mode of the binomial distribution is $(np+p)$.

Case 1: $(np+p)$ is fraction (n is even) mode = Integer part of $(np+p)$

Case 2: $(np+p)$ is an integer (n is odd) mode = $(np+p)$ & $(np+p-1)$

Q→ 5 coins are tossed 320 times & following observation are observed. Fit the binomial distr. or find the expected frequency.

x = No. of heads	0	1	2	3	4	5	
f(x)	7	40	100	110	60	3	$\Sigma f(x) = 320$
P(x)	1/32	5/32	10/32	10/32	5/32	1/32	
$f_e(x) = N \cdot P(x)$	10	50	100	100	50	10	$\Sigma f_e(x) = 320$

$N = 320$ $n = 5$ $p = 1/2$ $q = 1/2$ x = No. of heads = 0, 1, 2, 3, 4, 5.

$P(x) = {}^nC_x\, p^x q^{n-x} = {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$

$= {}^5C_x \left(\frac{1}{2^5}\right) = \frac{{}^5C_x}{32}$

$P(x=0) = \frac{{}^5C_0}{32} = \frac{1}{32} = P(x=5)$

$P(x=1) = \frac{{}^5C_1}{32} = \frac{5}{32} = P(x=4)$

$P(x=2) = \frac{{}^5C_2}{32} = \frac{10}{32} = P(x=3)$

jab probability fix nahi hoti tab use mean $= \frac{\Sigma f \cdot x}{\Sigma f} = np$.

then calc. p and q.

No. of mistakes	0	1	2	3	4	5
$f_o(x)$	150	80	45	40	3	2

$n = 5$

$N = 320$

p(x) of mistakes is not uniform.

$\frac{\Sigma f \cdot x}{\Sigma f} = np \Rightarrow \frac{80+90+120}{320} = 5 \times p$

$\Rightarrow \frac{312}{320} = 5 \times p \Rightarrow p = \frac{39}{200} = \frac{18.5}{100} = 0.185$

$q = 1 - 0.185 = 0.815$, x = no. of mistakes = 0, 1, 2, 3, 4, 5

$P(x) = {}^5C_x (0.185)^x (0.815)^{5-x}$

Q→ The sum of mean & variance of a B.D. is 15 and the sum of their squares is 117. Find the distribution.

$np + npq = 15$

$np(1+q) = 15$

$(np)^2 + (npq)^2 + 2 \cdot np \cdot npq = 225$

$117 + 2n^2p^2q = 225$

$2n^2p^2q = 108$

$n^2p^2q = 54$

$(np)^2 + (npq)^2 = 117$

$n^2p^2(1+q^2) = 117$

$\frac{(15-npq)^2}{(1+q)^2}(1+q^2) = 117$

$225 + 225q^2 = 117 + 117q^2 + 234q$

$108q^2 + 108q^2 - 234q = 0$

$54q^2 - 117q + 54 = 0$

$np(1+q) = 15$

$np = \frac{15}{1+q}$ ... ①

$n^2p^2q = 54$ ... ②

squaring ① & div by ②

$\frac{n^2p^2(1+q)^2}{n^2p^2q(1+q^2)} = \frac{225}{117}$

$q = 3/2, 2/3$ (discard $3/2$)

$p = 1/3$

$n = 27$

Q→ If 'x' follows a B.D. with mean 4 and variance 2. Find P(E) of $(|X-4| \le 2)$

$|X-4| \le 2$

$-2 \le X-4 \le 2$

$2 \le X \le 6$

$X^2 + 16 - 8X \le 2$

$X^2 - 8X + 14 \le 0$

$np = 4$ $\quad npq = 2$

$n \cdot 1/2 = 4$ $\quad 4 \cdot q = 2$

$n = 8$ $\quad q = 1/2$ $\quad p = 1/2$

$P(|X-4| \le 2) = {}^8C_r \left(\frac{1}{2}\right)^r \left(\frac{1}{2}\right)^{8-r}$ where $r = 2,3,4,5,6$.

Q→ The P(E) that Hari hits a target is 1/4. He fires 64 times. Find the expected no. of times he will hit the target and the variance.

$p = \frac{1}{4}$ $\quad q = \frac{3}{4}$ $\quad n = 64$.

expected no. of times = mean = $np = \frac{1}{4} \cdot 64 = 16$.

variance = $npq = 16 \cdot \frac{3}{4} = 12$.

Q→ The following is the frequency distribution of 128 throws of 7 coins according to the no. of heads.

No. of heads	0	1	2	3	4	5	6	7
Throws	7	6	19	35	30	23	7	1

Fit a B.D. What is the mean and Standard Deviation.

$N = 128$ $\quad n = 7$ $\quad p = 1/2$ $\quad q = 1/2$ $\quad x = 0,1,2,3,4,5,6,7$

$P(x) = {}^nC_x \, p^x q^{n-x}$

$P(x) = {}^7C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{7-x} = {}^7C_x \left(\frac{1}{2}\right)^7$

$P(x) = \frac{{}^7C_x}{128}$

P(x)	1/128	7/128	21/128	35/128	35/128	21/128	7/128	1/128
$f_e(x)$	1	7	21	35	35	21	7	1

Mean = $\frac{\Sigma f \cdot (x)}{\Sigma f(x)}$

$= \frac{6 + 38 + 105 + 120 + 115 + 42 + 7}{128}$

$= \frac{433}{128}$

Q→ Mean = 4 variance = 2

$P(|X-4| \le 2) = P(X=2) + P(X=3) + P(X=4) + P(X=5) + P(X=6)$

$= {}^8C_2 \left(\frac{1}{2}\right)^8 + {}^8C_3 \left(\frac{1}{2}\right)^8 + {}^8C_4 \left(\frac{1}{2}\right)^8 + {}^8C_5 \left(\frac{1}{2}\right)^8 + {}^8C_6 \left(\frac{1}{2}\right)^8$

$= \left(\frac{1}{2}\right)^8 \left[{}^8C_2 + {}^8C_3 + {}^8C_4 + {}^8C_5 + {}^8C_6\right]$

$= \left(\frac{1}{2}\right)^8 [28 + 56 + 56 + 28 + 70]$

$= \left(\frac{1}{2}\right)^8 [238] = \frac{238}{256}$

Q→ A dice is thrown 6 time. Getting an odd number is a success. What is P(E) of ⓐ getting 5 success ⓑ atmost 5 success.

$n=6 \quad p=3/6=1/2$

a) getting 5 success $= {}^6C_5 \left(\frac{1}{2}\right)^5 \left(\frac{1}{2}\right) = 6/64 = 3/32$

b) atmost 5 success $= P(x=0)+P(x=1)+\cdots\cdots+P(x=5)$
$= 1-P(x=6)$
$= 1-\left(\frac{1}{2}\right)^6 = 1-\frac{1}{64} = \frac{63}{64}$

Q→ The P(E) of a man hitting a target is 1/2. How many times must he fire so that the P(E) of hitting the target atleast once is more than 90%.

P(E) of hitting a target $= 1/2$.

P(E) of hitting target atleast once $> \frac{9}{10}$

$P(X \geq 1) > \frac{9}{10}$

$P(X=1)+P(X=2)+\cdots\cdots+P(X=n) > \frac{9}{10}$

$1-P(X=0) \Rightarrow 9/10$

$1-{}^nC_0\left(\frac{1}{2}\right)^0\left(\frac{1}{2}\right)^n > 9/10$

$1-\left(\frac{1}{2}\right)^n > \frac{9}{10}$

$1-\frac{9}{10} > \frac{1}{2^n}$

$\frac{1}{10} > \frac{1}{2^n}$

$2^n > 10 \qquad n=4,5,6,\cdots\cdots$ at least 4 times.

Q→ Find the Binomial Distribution whose mean is 3 & variance is 2.

$np=3 \qquad npq=2 \qquad np=3$

$p=1-2/3=1/3 \qquad 3\cdot q=2 \qquad n\cdot\frac{1}{3}=3$

$q=2/3 \qquad n=9$

$P(x) = {}^9C_x\left(\frac{1}{3}\right)^x\left(\frac{2}{3}\right)^{9-x}$

Poisson's Distribution:

Is applicable when no. of trial is large & prob. of success is small.

n = large $\quad p \to$ small

$P(x) = \frac{e^{-\lambda}(\lambda)^x}{x!} \qquad \lambda = np \qquad x = 0,1,2,\cdots\cdots,\infty$

Mean of B.D. = mean of P.D.

eg → No. of errors, No. of accidents on a road.

In a Poisson distribution, mean $= \lambda = np$

Variance $= \sigma^2 = \sqrt{\lambda} = \sqrt{np}$

Mode $= \lambda = np$

$E(x) = \sum_{x=0}^{\infty} x \cdot P(x) = \sum_{x=0}^{\infty} x \cdot \frac{e^{-\lambda}(\lambda)^x}{\lfloor x}$

$P(x) = \frac{e^{-\lambda}(\lambda)^x}{\lfloor x}$

$x = 0,1,2,\cdots\cdots\infty$

$\lambda = np$

$= e^{-\lambda}\sum_{x=0}^{\infty} x\cdot\frac{(\lambda)^x}{\lfloor x}$

$= e^{-\lambda}\sum_{x=0}^{\infty}\frac{x\cdot(\lambda)^x}{x(x-1)}$

$= e^{-\lambda}\left\{\frac{\lambda^0}{\lfloor 1} + \frac{\lambda^1}{\lfloor 0} + \frac{\lambda^2}{\lfloor 1} + \cdots\cdots+\infty\right\}$

$= e^{-\lambda}\left\{0+\lambda+\frac{\lambda^2}{\lfloor 1}+\cdots\cdots+\infty\right\}$

$= \lambda\cdot e^{-\lambda}\left\{1+\lambda+\frac{\lambda^2}{\lfloor 2}+\cdots\cdots+\infty\right\}$

$= \lambda\cdot e^{-\lambda}\cdot\{e^{\lambda}\} = \lambda$

Expected value of x $= E(x) = \lambda = np$

Q→ A car hire firm has 2 cars which it hires out day by day. The no. of demands for a car on each day is distributed as a Poisson distribution with mean 1.5. Cal. the proportion of the day (Probability) on which no car is used and proportion of the day when some demand is refused.

(no. of the day = frequency)

$P(x) = \frac{e^{-\lambda}(\lambda)^x}{\lfloor x} \qquad x =$ no. of demands $\qquad P(x) = \frac{e^{-\lambda}(\lambda)^x}{\lfloor x} = \frac{e^{-1.5}(1.5)^x}{\lfloor x}$

(i) no car is used

$P(x=0) = \frac{e^{-1.5}(1.5)^0}{\lfloor 0} = e^{-1.5}$

(ii) some demand refused

$P(x \geq 2) = P(x \geq 3) = P(3)+P(4)+\cdots\cdots+P(\cdot)\cdots\cdots$

$= 1-\{P(0)+P(1)+P(2)\}$

$= 1-\left\{\frac{e^{-1.5}(1.5)^0}{\lfloor 0} + \frac{e^{-1.5}(1.5)^1}{\lfloor 1} + \frac{e^{-1.5}(1.5)^2}{\lfloor 2}\right\}$

Q→ In a certain factory turning out razor blades, there are small chance of 1/500 for any blade to be defective. The blades are supplied in a packet of 10. Using appropriate distribution to calc. approximate no. of packets containing no defective, 1 defective, 2 defective, 3 defective blades in a consignment of 10,000 packets.

$N = 10,000 \quad n = 10 \quad p = 1/500 = 0.002$

x = no. of defective blades = 0, 1, 2, ——, 10

Here, Binomial & Poisson, both are fulfilling partially—

Binomial

$P(x) = {}^nC_x\, p^x q^{n-x}$

$P(x) = {}^{10}C_x\, (0.002)^x (0.998)^{10-x}$

$P(0) = 1\,(0.002)^0 (0.998)^{10-0}$

$P(0) = (0.998)^{10} = 0.98017$

$fe(x=0) = N \times P(x=0)$
$= 10,000 \times 0.98017$
$= 9801.7 = 9802$

$P(x=1) = {}^{10}C_1\, (0.002)^1 (0.998)^{10-1}$
$= 10\,(0.002)(0.998)^9$
$= 0.019642$

$fe(x=1) = 196.42 = 196$

$P(x=2) = {}^{10}C_2\, (0.002)^2 (0.998)^{10-2}$
$= {}^{10}C_2\, (0.002)^2 (0.998)^8$
$= 0.000177$

$fe(x=2) = 1.77 \simeq 2$

$P(x=3) = {}^{10}C_3\, (0.002)^3 (0.998)^{10-3}$
$= {}^{10}C_3\, (0.002)^3 (0.998)^7$
$= 0.0000008$

$fe(x=3) = 0.0008 \simeq 0.$

Poisson

$P(x) = \dfrac{e^{-\lambda}(\lambda)^x}{\lfloor x}$

$np = 10 \times 0.002$
$= 0.02$

$P(x=0) = \dfrac{e^{-0.02}(0.02)^0}{\lfloor 0} = e^{-0.02}$
$= 0.98019$

$fe(x=0) = N \times P(X=0)$
$= 9801.9 = 9802$

$P(x=1) = \dfrac{e^{-0.02}(0.02)^1}{\lfloor 1}$
$= e^{-0.02} \times 0.02$
$= 0.019603$

$fe(x=1) = 196.03 = 196$

$P(x=2) = \dfrac{e^{-0.02}(0.02)^2}{\lfloor 2}$
$= 0.000196$

$fe(x=2) = 1.96 \simeq 2$

$P(x=3) = \dfrac{e^{-0.02}(0.02)^3}{\lfloor 3}$
$= 0.000007$

$fe(x=3) = 0.007$
$= 0.$

Probability ~~distribution~~ of defective bolt decreases.

Q→ Assuming that typing mistake per page committed by a typist follows a Poisson Distribution. Find the Expected frequency for the following dist^n of typing mistakes.

No. of mistake per page	0	1	2	3	4	5
No. of pages	40	30	20	15	10	5

$X \to$ no. of mistakes per page

$p(x) = \dfrac{e^{-\lambda}\lambda^x}{x!}$

$\lambda = np$ = mean

X	0	1	2	3	4	5
observed f_o	40	30	20	15	10	5
$fe(x) = N \cdot f(x)$						

$\text{Mean} = \dfrac{\sum fx}{\sum f} = \dfrac{30+40+45+40+25}{120} = \dfrac{180}{120} = \dfrac{3}{2} = 1.5$

$\lambda = 1.5$

$p(x=0) = \dfrac{e^{-1.5}(+1.5)^0}{0!} = e^{-1.5} = 0.2231$

$p(x=1) = e^{-1.5}(1.5)^1 = 0.3346$

$p(x=2) = 0.25102$

$p(x=3) = \dfrac{e^{-1.5}(1.5)^3}{6} = 0.125493$

$p(x=4) = \dfrac{e^{-1.5}(1.5)^4}{4!} = 0.04706$

$p(x=5) = \dfrac{e^{-1.5}(1.5)^5}{5!} = 0.0141180$

X	f_o	$f(x)$	$fe(x) = N \cdot p(x)$
0	40	0.2231	$26.772 = 27$
1	30	0.3346	$40.152 = 40$
2	20	0.2509875	$30.1185 = 30$
3	15	0.125493	$15.05916 = 15$
4	10	0.04706	$5.6472 = 6$
5	5	0.0141180	$1.6946 = 2$
	120		120

Random Variable & Prob. distribution

R.V. is a variable which assign some ~~numerical~~ random value.

They are of 2 types ① discrete R.V. $x = 0, 1, 2, \ldots n$ ② continuous R.V. $x \in [0, 1]$

$e^{\infty} = \infty$

$e^{-\infty} = 0$

Probability Distribution → If $p(x)$ is a real value func. which satisfy following condition: (i) $p(x) \geq 0 \ \forall x$ then $p(x)$ is called prob. distn.

(ii) $\sum_x p(x) = 1$

$\Sigma p(x) = \sum_{x=0}^{\infty} \dfrac{e^{-\lambda}(\lambda)^x}{\lfloor x}$

$= e^{-\lambda} \sum_{x=0}^{\infty} \dfrac{(\lambda)^x}{\lfloor x}$

$= e^{-\lambda} \left\{ \dfrac{1}{0} + \dfrac{\lambda}{1} + \dfrac{\lambda^2}{2} + \cdots + \dfrac{1}{\infty} + \cdots \infty \right\}$

$= e^{-\lambda} e^{\lambda} = 1$

Q → If $P(x)$ represent $\frac{(\lambda)^x}{x}$; $x=1,2,\dots,\infty$ is a prob. distribution then what is the value of λ?

$$P(x)=\frac{(\lambda)^x}{x} \quad ; x=1,2,\dots,\infty$$

$$\left\{\frac{\lambda}{1}+\frac{\lambda^2}{2}+\dots+\infty\right\}=1$$

$$\left\{1+\frac{\lambda}{1}+\frac{\lambda^2}{2}+\dots-\infty\right\}=1+1$$

$$e^{\lambda}=2$$

$$\lambda=\log_e 2$$

* $[0,10]\to x\in[0,10]$ then find Σx^2.

Its not possible as there are infinite terms. So we integrate it as — "Definite Integral has a limit of sum"

It should satisfy —

(i) $P(x)\geq 0$

(ii) $\Sigma P(x)=1$

given $\Rightarrow \sum_{x=1}^{\infty}\frac{(\lambda)^x}{x}=1$

$\Rightarrow \lambda\sum_{x=1}^{\infty}\frac{1}{x}=1$

$\Rightarrow \lambda\left\{\frac{1}{1}+\frac{1}{2}+\dots+\infty\right\}=1$

$\Rightarrow \lambda\left\{\left(1+\frac{1}{1}+\frac{1}{2}+\dots+\infty\right)-1\right\}=1$

$\Rightarrow \lambda(e^{\lambda}-1)=1$

$\lambda=\frac{1}{e^{\lambda}-1}$

$\frac{1}{e-1}\geq 0$, $\frac{1}{1-e}<0$

Probability Dynamic function.

A func. $y=f(x)$ is a real valued func. is defined as in $[a,b]$ $a\leq x\leq b$, is P.d.f. when (i) $f(x)\geq 0\ \forall x$ (ii) $\int_a^b f(x)dx=1$

eg1→ $f(x)=\sin x$, $x\in(0,\frac{\pi}{2})$

(i) $f(x)\geq 0\ \forall x$

(ii) $\int_0^{\pi/2}\sin x=[-\cos x]_0^{\pi/2}=[-0+1]=1$.

eg2→ $f(x)=\sin x$, $x\in[0,\pi]\to[-\cos x]_0^{\pi}=1+1=2$.

If a func. $y=f(x)$ is defined as $f(x)=K(x-x^2)$; $x\in[0,1]$ is a p.d.f. then (i) find the value of K.

(ii) find the mean of $f(x)$.

(iii) Mode of $f(x)$

(iv) Median of $f(x)$

(v) Is distribution symmetric — ?

$\therefore E(x)=\int x\cdot f(x)dx$

$\sigma^2=\int(x-\bar{x})^2 f(x)dx$.

→ Given that $f(x)$ p.d.f. $\begin{cases}f(x)\geq 0\\ \int f(x)dx=1\end{cases}$

$$\int_0^1 f(x)dx=1 \Rightarrow \int_0^1 K(x-x^2)dx=1 = K\left(\frac{x^2}{2}-\frac{x^3}{3}\right)=1$$

$\boxed{K=6}$ $\qquad f(x)=6(x-x^2)$.

(ii) Mean $=E(x)$

$=\int x\cdot f(x)dx$

$=\int x\cdot 6\cdot(x-x^2)dx$.

$6\left(\frac{x^3}{3}-\frac{x^4}{4}\right)_0^1=\frac{1}{2}$.

(iii) Mode → that value of x for which $p(x)$ or $f(x)$ is maximum. Since there are 3 values, so mode can't be obtained directly.

using Maxima and Minima, $y=f(x)$ $\left(\frac{d^2y}{dx^2}\right)_{x=b}=+ve$

$f(x)=6(x-x^2)$ $\qquad \left(\frac{d^2y}{dx^2}\right)_{x=a}=-ve$ (concave)

$\frac{dy}{dx}=0$ (stationary value) $\qquad \frac{d^2y}{dx^2}$

$\frac{d^2y}{dx^2}=+ve$ (minima)

f(x) is said to be global maxima at $x=a$ if it is maximum in interval.

$f(x)=6(x-x^2)$ $x\in[0,1]$

$\frac{dy}{dx}=-12<0$ So it implies that $f(x)$ is maximum at $x=1/2$.

Hence mode $=1/2$

(iv) Median → $\int_0^1 f(x)dx=1=\int_0^{Me}f(x)dx=\int_{Me}^1 f(x)dx=\frac{1}{2}$

$\Rightarrow \int_0^{Me}f(x)dx=\frac{1}{2}$ — Me = ?

$\Rightarrow \int_0^{Me}6(x-x^2)dx=\frac{1}{2}\Rightarrow 6\left(\frac{x^2}{2}-\frac{x^3}{3}\right)_0^{Me}=\frac{1}{2}$

$\Rightarrow 6\left(\frac{3Me^2-2Me^3}{6}\right)=\frac{1}{2}$

$\Rightarrow 4Me^3-6Me^2+1=0$

$Me=\frac{1}{2}, -0.33, 1.33$ — out of bounds

∴ Median $=1/2$

parameters of statistical
① Mean ② Standard deviation

(v)

x	0	0.1	0.2		0.5	0.8	0.9	1
	0	0.54	0.96		1.5	0.96	0.54	0

$=x$

If mean = mode = median then curve is always symmetric about x.

(vi) Standard Deviation (σ)

$\sigma^2=\int_0^1(x-\bar{x})^2 f(x)dx$. ; $\bar{x}=\text{mean}=E(x)=\frac{1}{2}$

$\sigma^2=\int_0^1\left(x-\frac{1}{2}\right)^2 6(x-x^2)dx$.

$\sigma^2=\int_0^1\left(x^2+\frac{1}{4}-x\right)(6x-6x^2)dx=\int_0^1\left(6x^3+\frac{3}{2}x-6x^2-6x^4-\frac{3}{2}x^2+6x^3\right)dx$

$\sigma^2=\int_0^1\left(12x^3-6x^4-\frac{15}{2}x^2+\frac{3}{2}x\right)dx$

$\sigma^2=\left(\frac{12x^4}{4}-\frac{6x^5}{5}-\frac{15x^3}{2\times 3}+\frac{3}{4}x^2\right)_0^1=3-\frac{6}{5}-\frac{5}{2}+\frac{3}{4}=\frac{60-24-50+15}{20}$

$\sigma^2=1/20$ $\qquad \sigma=1/\sqrt{20}$

Normal distribution → is a continuous distri. where prob. density func. of the normal distri. is represented as

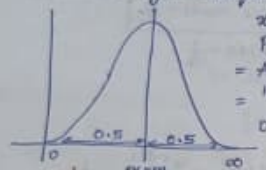
$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{\frac{-(x-m)^2}{2\sigma^2}}$$

m = mean = median = mode

σ = standard deviation.

$x \in (0,\infty)$

It is a symmetric func.



$x=1$ to 10

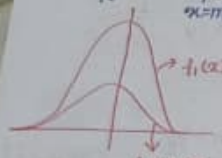
$P(1\le x\le 10) = A(1\le x\le 10) = \int_0^{10} f(x)\,dx$

$$y=f(x)=\frac{1}{\sigma\sqrt{2\pi}}e^{\frac{-(x-m)^2}{2\sigma^2}}$$

$$z=\frac{x-m}{\sigma}$$

$$y=f(z)=\frac{1}{\sigma\sqrt{2\pi}}e^{\frac{-z^2}{2}} \quad \to z\in(-\infty,\infty)$$

$x=m$	$z=0$
$x>m$	$z>0$
$x<m$	$z<0$



$\sigma_1 < \sigma_2$

*The more the ~~dense~~ spread, the greater the S.D.

Given that $\phi(z)=\int_0^z f(z)\,dz$, $\phi(1)=0.34134$.

Q→ The scores made by candidates in a certain test are normally distri. with mean 500 and S.D. 100. What % of candidate recieve the score between 400 & 600. $m=500$ S.D. $=100$

$P(400<x<600)$

$P(-1<z<1)$

$= A(-1<z<1)$

$= A(-1<z<0) + A(0<z<1)$

$= \phi(-1)+\phi(1)$

$= 2A(0<z<1)$

$= 2\phi(1)$

$= 2\times 0.34134$

$= 0.68268$

$$z=\frac{x-m}{\sigma}=\frac{x-500}{100}$$

$x_1=400$, $z=\frac{400-500}{100}=-1$

$x_2=600$, $z=\frac{600-500}{100}=1$

Given $A(0<z<1)=0.34134$

percentage candidate $= 0.68268\times 100$

$= 68.268\%$

Q→ The lifetime of certain type of batteries has a mean of 400 & S.D = 45 hours. Assuming a distrib. of lifetime to be normal then find

(i.) % of battery with a lifeline of atleast 470 hours.

(ii.) Propation of battery with lifetime b/w 385 & 415 hours. $P(385<x<415)$

(iii.) Minimum lifetime of the best 5% battery. (more than 5%)

(i.) % $P(x>470)$

$m=400$ $\sigma=45$

$$z=\frac{x-m}{\sigma}=\frac{x-400}{45}$$

Here, $x=470 \longrightarrow z=\frac{470-400}{45}=1.55\ldots\approx 1.56$

$P(x>470)=P(z>1.56)=A(z>1.56)$

$\phi(z)=\int_0^z f(z)\,dz$ $\qquad \phi(1.56)=0.4406$

$A(z>1.56)=0.5-A(0<z<1.56)$

$=0.5-0.44406=0.0594=5.94\%$

(ii.) $P(385<x<415)$

$x=385 \longrightarrow z=\frac{385-400}{45}=-0.33$

$x=415 \longrightarrow z=\frac{415-400}{45}=0.33$

$P(385<x<415)=P(-0.33<z<0.33)$

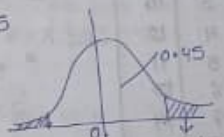
Given $\phi(0.33)=0.1293$

$=(-0.33<z<0)+(0<z<0.33)$

$=0.1293+0.1293$

$=0.2586$

(iii.) Given that $\phi(1.65)=0.45 \longrightarrow z=1.65$



$1.65=\frac{x-400}{45}$

$x=474.25\approx 474$

Q→ The marks of the students are normally distributed. 10% get more than 75 marks. 20% get less than 40 marks. Then find the mean & S.D. of distribution.

Given that

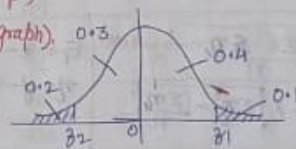
x	0.84	1.28	2.0
A	0.2995	0.3997	0.4772

x	A
$x>75$	10% = 0.10
$x\le 40$	20% = 0.20

$x_1=75 \longrightarrow z_1=\frac{75-m}{\sigma}$ $\qquad z_2=\frac{40-m}{\sigma}$

go > (right in graph)

< (go left in graph)



$\phi(0.84)=0.3$ $\quad \phi(z_1)=0.340$

$\phi(1.28)=0.4$ $\quad \phi(z_2)=0.30$

$z_1=1.28=1.28$

$z_2=0.84=-0.84$

$z_1=\frac{75-m}{\sigma}=1.28$

$75-m=1.28\sigma$

$z_2=\frac{40-m}{\sigma}=-0.84$

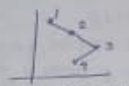
$40-m=-0.84\sigma$

$m=53.88$ $\qquad \sigma=16.5$

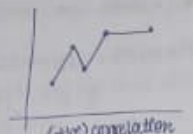
* Correlation and Relation

x	y
x_1	$y_1 \longrightarrow 1=(x_1, y_1)$
x_2	$y_2 \longrightarrow 2=(x_2, y_2)$
x_3	$y_3 \longrightarrow 3=(x_3, y_3)$
-	-

$r = +1$
$r = -1$
$-1 \le r \le 1$



(-ve) correlation



(+ve) correlation

* Find the Karl Pearson's power of correlation:

Calc. the coff. ~~of correlation~~ coefficient of the correlation of following data:

x	y
1	9
2	8
3	10
4	12
5	11
6	13
7	14
8	16
9	15
$\Sigma x = 45$	$\Sigma y = 108$

$$r = \frac{\Sigma XY}{\sqrt{\Sigma X^2}\sqrt{\Sigma Y^2}}$$

$Y = y - \bar{y}$
$X = x - \bar{x}$

$\bar{x} = \frac{45}{9} = 5$, $\bar{y} = \frac{108}{9} = 12$

Calc. $X = x - \bar{x}$ & $Y = y - \bar{y}$ for each.

$\Sigma X = 0$, $\Sigma Y = 0$

$\Sigma X^2 = 60$ $\Sigma Y^2 = 60$

$\Sigma X \cdot Y = 57$.

$r = \frac{57}{60 \times 60} = 0.95$.

used in:- feature selection.

$\bar{x} = \frac{\Sigma x}{n}$

$\bar{y} = \frac{\Sigma y}{n}$

X	Y
+4	-3
-4	-4
-3	-1
-2	-2
-1	0
0	-1
1	1
2	2
3	4
4	3
$\Sigma =$ 0	0

Q→

x	y
77	35
54	58
27	60
52	40
14	50
35	40
90	35
25	36
56	34
60	42

Now we solve the problem by

$$r = \frac{\Sigma \xi \eta - \frac{\Sigma \xi \cdot \Sigma \eta}{n}}{\sqrt{\Sigma \xi^2 - \frac{(\Sigma \xi)^2}{n}} \times \sqrt{\Sigma \eta^2 - \frac{(\Sigma \eta)^2}{n}}}$$

(reverse)

$\xi = x - Ax$
$\eta = y - Ay$.

$$r = \frac{81 - \frac{350 \times 30}{10}}{\sqrt{17400 - \frac{(350)^2}{10}} \times \sqrt{930 - \frac{(30)^2}{10}}}$$

$$r = \frac{-969}{71.76 \times 28.98} = -0.46588$$

≈ 0.46.

x	y	$\xi_i = x - Ax$ $= x - 14$	$\eta = y - Ay$ $= y - 40$	$\xi\eta$	ξ^2	η^2
77	35	63	-5	-315	3969	25
54	58	40	18	720	1600	324
27	60	13	20	260	169	400
52	40	38	0	0	1444	0
Ax → (14)	50	0	10	0	0	100
35	(40) Ay	21	0	0	441	0
90	35	76	-5	-380	5776	25
25	36	11	-4	-44	121	16
56	34	42	-6	-252	1764	36
60	42	46	2	92	2116	4
	$\Sigma =$	350	30	81	17,400	930

x's

* Spearman Rank Correlation:

$-1 \le R \le +1$.

$$R = 1 - \frac{6 \Sigma D^2}{n(n^2-1)}$$

$\therefore D^2$ = diff. b/w ranks
= Rank of x − Rank of y
n = no. of items in sample space.

If rank of 2 are same then
$R = x - \frac{\text{rank 1} + \text{rank 2}}{2}$

Q→ Find the Rank coefficient of the following data:
10 students get the following % of marks in mathematics & physics

Sno.	Maths (x)	Physics (y)	Rank of x	Rank of y	D = (Rank of x − Rank of y)	D^2
1	8	84	10	3	7	49
2	36	51	7	8	-1	1
3	98	91	1	1	0	0
4	25	60	9	6	3	9
5	75	68	4	4	0	0
6	82	62	3	5	-2	4
7	92	86	2	2	0	0
8	62	58	6	7	-1	1
9	65	35	5	10	-5	25
10	35	49	8	9	-1	1
						$\Sigma D^2 = 90$

$n=10$
$\Sigma D^2=90$

$$R=1-\frac{6\Sigma D^2}{n(n^2-1)}$$

$$R=1-\frac{6\times90}{10(100-1)}=\frac{540}{990}$$

$$\boxed{R=0.45}$$

Q. 10 competitor in a beauty contest are ranked by following order :-

S No	I (X)	II (Y)	III (Z)	DXY	$(DXY)^2$	$(DYZ)^2$	$D(XZ)^2$
1	1	3	6	-2	4	9	25
2	6	5	4	-1	1	1	4
3	5	8	9	-3	9	1	16
4	10	4	8	6	36	16	4
5	3	7	1	-4	16	36	4
6	2	10	2	-8	64	64	0
7	4	2	3	2	4	1	1
8	9	1	10	8	64	81	1
9	7	6	5	1	1	1	4
10	8	9	7	1	1	4	1
					$\Sigma=200$	$\Sigma=214$	$\Sigma=60$

$$R_{XY}=1-\frac{6\Sigma DXY^2}{n(n^2-1)}=1-\frac{6\times200}{10\times99}$$

$$\boxed{R_{XY}=-0.21}$$

$$R_{YZ}=1-\frac{6\Sigma(DYZ)^2}{n(n^2-1)}=1-\frac{6\times214}{10\times99}$$

$$\boxed{R_{YZ}=-0.99}$$

$$R_{XZ}=1-\frac{6\Sigma D(XZ)^2}{n(n^2-1)}=1-\frac{6\times60}{10\times99}=\boxed{0.6}$$

use the correlation coffi. to determine which pair of judge have the nearest approach to the common

So X & Z have the nearest common approach.

- outlier – does not affect median but changes mean significantly.
→ more data, outliers = more data overreading.

* Regression (real value).
(prediction and forecasting). (Error cannot be found.)

Regression Analysis (linear regression)

In R.A, we define 2 line – first line is Y on X:

b_{yx} = Regression coefficient of Y on X

$$\boxed{(y-\bar{y})=\frac{r\sigma_y}{\sigma_x}(x-\bar{x})}$$

x → Given
Y → ?

$\bar{x}$ = mean of x
σ_x = S.D. of x
$\bar{y}$ = mean of y
σ_y = S.D. of y
r = correlation coefficient.

b_{xy} = Regression coff. of X on Y.

• X on Y:- $(x-\bar{x})=\frac{r\sigma_x}{\sigma_y}(y-\bar{y})$

Y → Given
X → ?

2 line isliye bnti hai taaki pair find ka paati hai ek baar vertical, ek baar horizontal.

* Angle b/w two regression line :

$y_1=m_1x+c_1$ — ①
$y_2=m_2x+c_2$ — ②

$$\tan\theta=\frac{m_1\sim m_2}{1+m_1\cdot m_2}$$

→ We consider (-ve) sign, if both regression coff. are +ve.
→ If one is +ve & one is -ve, no becomes imaginary so its not possible.

$$\begin{cases} b_{xy}\times b_{yx}=\frac{r\sigma_y}{\sigma_x}\times\frac{r\sigma_x}{\sigma_y}=r^2. \\ \boxed{r=\pm\sqrt{b_{yx}\times b_{xy}}} \end{cases}$$

For line Y on X :

$$(y-\bar{y})=\frac{r\sigma_y}{\sigma_x}(x-\bar{x})$$

$$y-\bar{y}=\frac{r\sigma_y}{\sigma_x}x-\frac{r\sigma_y}{\sigma_x}\bar{x}$$

$$y=\frac{r\sigma_y}{\sigma_x}x+\boxed{\left(\bar{y}-\frac{r\sigma_y}{\sigma_x}\bar{x}\right)}\quad\therefore y=mx+c_1$$

→ c

$$\boxed{m_1=\frac{r\sigma_y}{\sigma_x}}$$

For line X on Y:

~~$x-\bar{x}=\frac{r\sigma_x}{\sigma_y}(y-\bar{y})$~~

$$x-\bar{x}=\frac{r\sigma_x}{\sigma_y}(y-\bar{y})$$

$$\Rightarrow y-\bar{y}=\frac{1}{r}\frac{\sigma_y}{\sigma_x}(x-\bar{x})$$

$$y=\frac{1}{r}\frac{\sigma_y}{\sigma_x}x-\frac{1}{r}\frac{\sigma_y}{\sigma_x}\bar{x}+\bar{y}$$

$$y=\left(\frac{1}{r}\frac{\sigma_y}{\sigma_x}\right)x+\bar{y}\Big($$

$$\boxed{m_2=\frac{1}{r}\frac{\sigma_y}{\sigma_x}}$$

$$\tan\theta=\frac{m_1\sim m_2}{1+m_1m_2}=\frac{m_2-m_1}{1+m_1\cdot m_2}=\frac{\frac{1}{r}\frac{\sigma_y}{\sigma_x}-r\frac{\sigma_y}{\sigma_x}}{1+\frac{1}{r}\frac{\sigma_y}{\sigma_x}\cdot r\frac{\sigma_y}{\sigma_x}}$$

$\tan\theta =$

$$\boxed{\tan\theta = \frac{1-r^2}{r}\,\frac{\sigma_x\cdot\sigma_y}{\sigma_x^2+\sigma_y^2}}$$

If $\theta = 0$, (both line coincides)

$$\tan\theta = \frac{1-r^2}{r}\left(\frac{\sigma_x\cdot\sigma_y}{\sigma_x^2+\sigma_y^2}\right)$$

$$0 = 1-r^2$$

$$\boxed{r = \pm 1}$$

• If $r=0$, lines are totally different.

$$\theta = 90^\circ \Rightarrow \tan 90^\circ = \left(\frac{1-r^2}{r}\right)\cdot\frac{\sigma_x\cdot\sigma_y}{\sigma_x^2+\sigma_y^2}$$

$$\infty = \left(\frac{1-r^2}{r}\right)\cdot\frac{\sigma_x\sigma_y}{\sigma_x^2+\sigma_y^2}.$$

Q→ In a laboratory experiment on correlation research study, the eqⁿ to the 2 regression lines were found to be—

$$2x-y+1=0 \quad \text{(i) and}$$

$$3x-2y+7=0 \quad \text{(ii)}$$

Find the mean of x, mean of y. Also find regression coefficients & correlation coefficients. Also, if the variance of x is 9, find the standard deviation of one?

First assuming that (i) is Y on X & (ii) is X on Y.

(Y on X): $y = 2x+1$

$$(y-\bar{y}) = r\frac{\sigma_y}{\sigma_x}(x-\bar{x})$$

$$b_{yx} = r\frac{\sigma_y}{\sigma_x} = 2.$$

(X on Y): $3x = 2y-7$. :

$$x = \frac{2}{3}y + \left(-\frac{7}{3}\right)$$

$$x-\bar{x} = r\frac{\sigma_x}{\sigma_y}(y-\bar{y})$$

$$b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{2}{3}$$

Now $r = \pm\sqrt{2\times\frac{2}{3}} = \frac{2}{\sqrt{3}} > 1$ (not possible).

So the assumption is wrong.

Now assuming that (i) is X on Y and (ii) is Y on X.

X on Y: $2x = y-1$

$$x = \frac{1}{2}y + \left(\frac{1}{2}\right)$$

$$b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{1}{2}$$

Y on X: $2y = 3x+7$.

$$y = \frac{3}{2}x + \frac{7}{2}.$$

$$b_{yx} = r\frac{\sigma_y}{\sigma_x} = 3/2.$$

$$r = \pm\sqrt{b_{xy}\times b_{yx}} = \pm\sqrt{\frac{1}{2}\times\frac{3}{2}} = \pm\frac{\sqrt{3}}{2}$$

Variance $(\sigma_x)^2 = 9$

$\sigma_x = 3$.

$r = \frac{\sqrt{3}}{2}$.

jiska variance kam hota hai wo jada dense hota hai

$$r\frac{\sigma_x}{\sigma_y} = \frac{1}{2} \Rightarrow \frac{\sqrt{3}}{2}\times\frac{3}{\sigma_y} = \frac{1}{2}$$

$$\sigma_y = 3\sqrt{3}.$$

Intersecting point of both the lines is their mean → $P(\bar{x}, \bar{y})$.

$$2x-y+1=0$$
$$3x-2y+7=0$$

$$6x-3y+3=0$$
$$6x-4y+14=0$$
$$+\quad -$$
$$y-11=0 \qquad \boxed{y=11}$$

$$2x-11+1=0$$
$$2x-10=0$$
$$\boxed{x=5}$$

Q→ Lines of regression is defined as—

$$2x+3y=7 \quad \text{(i)}, \qquad 5x+4y=9 \quad \text{(ii)}$$

(i) Find the $\bar{x}$ & $\bar{y}$. (iii) value of r?

(ii) Find b_{xy} & b_{yx}.

$$10x+15y=35$$
$$10x+8y=18$$
$$-\quad -$$
$$7y=17 \qquad \boxed{y=\frac{17}{7}}$$

$$2x+3\times\frac{17}{7} = 7.$$

$$14x+51=49$$

$$\bar{x} = \frac{-2}{14} = \frac{-1}{7}$$

assuming (i) to be Y on X.

$$y-\bar{y} = r\frac{\sigma_y}{\sigma_x}(x-\bar{x})$$

$$3y = -2x+7$$

$$y = \left(\frac{-2}{3}\right)x+7.$$

$$b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{-2}{3}$$

(ii) to be X on Y.

$$x-\bar{x} = r\frac{\sigma_x}{\sigma_y}(y-\bar{y})$$

$$5x = -4y+9$$

$$x = -\frac{4}{5}y+9/5$$

$$b_{xy} = r\frac{\sigma_x}{\sigma_y} = -4/5.$$

$$r = \pm\sqrt{\frac{-2}{3}\times\frac{-4}{5}} = \pm\sqrt{\frac{8}{15}} = \pm\frac{2\sqrt{2}}{\sqrt{15}} = \frac{-2\sqrt{2}}{\sqrt{15}}$$

(since both b_{xy} & b_{yx} are -ve).